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MATERIAL BIN APPARATUS AND ADHESIVE APPLICATION DEVICE

Abstract

A material bin apparatus includes a base and a plurality of material bins. An accommodating space is formed in each material bins. The accommodating space is configured for stacking and storing a plurality of materials. The material bin includes a fixed member and a movable member. The fixed member is fixed to the base. The movable member is movably disposed on the base and defines the accommodating space with the fixed member. The movable members of at least some of the material bins can synchronously move away from or approach the fixed members.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/CN2023/136171, filed on Dec. 4, 2023, which claims priority to Chinese Patent Application No. 202311289703.6, filed on Oct. 8, 2023 and entitled “MATERIAL BIN APPARATUS AND ADHESIVE APPLICATION DEVICE”, which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

[0002] This disclosure relates to the field of battery technologies, and specifically, to a material bin apparatus and an adhesive application device.

BACKGROUND

[0003] The application of new energy batteries in both daily life and industry is becoming increasingly widespread. For example, new energy vehicles equipped with batteries have been widely used, and batteries are also increasingly applied in the field of energy storage, among others.

[0004] In the production of a battery pack, an automatic adhesive application machine can be used to attach a double-sided adhesive tape to a surface of a battery cell. During the production process, a material bin apparatus configured to store a double-sided adhesive tape cannot accommodate a variety of types of double-sided adhesive tapes, resulting in long tooling changeover times and reduced production efficiency.

SUMMARY

[0005] To solve the above technical problems, this disclosure provides a material bin apparatus and an adhesive application device with good compatibility and high production efficiency. This disclosure is implemented through the following technical solutions.

[0006] According to a first aspect, this disclosure provides a material bin apparatus, the material bin apparatus including: [0007] a base; and [0008] a plurality of material bins, where an accommodating space is formed in each of the material bins, and the accommodating space is configured for stacking and storing a plurality of materials; and [0009] the material bin includes a fixed member and a movable member, where the fixed member is fixed to the base; the movable member is movably disposed on the base and defines the accommodating space with the fixed member; and the movable members of at least some of the material bins can synchronously move away from or approach the fixed members.

[0010] Thus, the plurality of material bins can be configured to simultaneously feed, improving the feeding efficiency, thereby improving the production efficiency. Moreover, different types of materials (such as strip materials and plate materials) can be accommodated by adjusting a position of the movable member, thereby improving the compatibility of the material bin apparatus. In addition, the movable members of at least some of the material bins can synchronously move away from or approach the fixed members. That is, the movable members of the plurality of material bins can be simultaneously adjusted. This is conducive to achieving quick changeover, thereby improving the production efficiency.

[0011] In some embodiments, the material bin apparatus includes a plurality of material zones, each of the material zones includes a plurality of material bins, and each of the material zones has a feeding state and a preparation state.

[0012] The material bin apparatus is provided with the plurality of material zones. During the use

of the material bin apparatus, when the material in the feeding state of the material zone is used up, the material zone in the feeding state is switched to the preparation state for material supplement, and the material zone in the preparation state is synchronously switched to the feeding state, thus allowing for feeding without shutdown and material change without shutdown, avoiding the problem of waiting for material change, and improving the processing efficiency.

[0013] In some embodiments, the base can rotate to switch each of the material zones between the feeding state and the preparation state.

[0014] That is, the base is controlled to rotate so as to switch each material zone between the feeding state and the preparation state. The structure for rotation of the base is simple, and a control method is simple and efficient.

[0015] In some embodiments, the material bin apparatus includes a turntable, the base is disposed on the turntable, and the turntable rotates to drive the base to rotate.

[0016] That is, the base is controlled to rotate so as to switch each material zone between the feeding state and the preparation state. The structure for rotation of the base is simple, and the control method is simple and efficient.

[0017] In some embodiments, the material bin apparatus includes a spacer disposed on the base; and two material zones are provided, and the two material zones are symmetrically disposed on two sides of the spacer.

[0018] The material bin apparatus includes a spacer disposed on the base. The spacer can be configured to separate different material zones. This is conducive to avoiding the influence between adjacent material zones to some extent, and ensuring the reliability of feeding and preparation in the material bin apparatus. In addition, the spacer can also be used as a positioning reference or for orientation, improving the convenience of using the material bin apparatus.

[0019] Two material zones are provided. Moreover, the two material zones are symmetrically disposed on two sides of the spacer. The spacer is configured to separate the two material zones, with one material zone being in use and the other in preparation. The provision of the spacer is conducive to avoiding the influence between feeding and preparation to some extent.

[0020] In some embodiments, the material bins of each of the material zones are arranged in a straight line. The movable member includes a first movable member. The material bin apparatus includes a drive mechanism, and the drive mechanism can be configured to synchronously drive first movable members of each of the material zones to reciprocate along a first direction.

[0021] Thus, the movement of all the first movable members of each material zone can be simultaneously controlled through the drive mechanism, so as to achieve the simultaneous changeover of all the material bins of each material zone, improving the changeover efficiency, and allowing for minute-level quick changeover, thereby improving the production efficiency.

[0022] In some embodiments, the movable member includes a second movable member. The drive mechanism can be configured to synchronously drive the second movable members of each of the material zones to reciprocate along a second direction, and the second direction intersecting the first direction.

[0023] Thus, the movement of all the second movable members of each material zone can be simultaneously controlled through the drive mechanism, so as to achieve the simultaneous changeover of all the material bins of each material zone, improving the changeover efficiency, and allowing for minute-level quick changeover, thereby improving the production efficiency.

[0024] In some embodiments, the first direction is an arrangement direction of each of the material bins in each of the material zones.

[0025] Thus, this is conducive to simultaneously controlling all the first movable members of each material zone to move along the first direction.

[0026] In some embodiments, the drive mechanism includes a first lead screw and a plurality of first nuts mating with the first lead screw; a first mounting groove and first sliding rail extending along the first direction are provided on the base; the first lead screw is rotatably disposed in the

first mounting groove; the first nut is fixed to the bottom of each of the first movable members; the first lead screw rotates to drive each of the first nuts to reciprocate along the first direction; and each of the first nuts drives each of the first movable members to reciprocate along the first sliding rail.

[0027] One first lead screw is provided, each first movable member is connected and mated with the first lead screw through the first nut, thereby controlling the movement of all the first movable members of each material zone, achieving simultaneously changeover of all the material bins in each material zone, improving the changeover efficiency, and allowing for minute-level quick changeover, thereby improving the production efficiency. The structure of the drive mechanism is simple and reliable.

[0028] In some embodiments, the drive mechanism includes a second lead screw, a connecting member, and a second nut mating with the second lead screw; a second mounting groove and second sliding rail extending along the second direction are provided on the base; the second lead screw is rotatably disposed in the second mounting groove; the bottom of each of the second movable members and the second nut are both fixed on the connecting member; and the second lead screw rotates to drive the second nut to reciprocate along the second direction, so that the connecting member drives each of the second movable members to reciprocate along the second sliding rail.

[0029] One second lead screw is provided, and each second movable member is connected and mated with the second lead screw through the second nut and the connecting member, thereby controlling the movement of all the second movable members of each material zone, achieving simultaneously changeover of all the material bins in each material zone, improving the changeover efficiency, and allowing for minute-level quick changeover, thereby improving the production efficiency. The structure of the drive mechanism is simple and reliable.

[0030] In some embodiments, the fixed member includes a first fixed member and a second fixed member, the first fixed member is disposed relative to the first movable member, and the second fixed member is disposed relative to the second movable member.

[0031] The material bin includes the first fixed member, the second fixed member, the first movable member, and the second movable member. The double-sided adhesive tape can be limited through the first fixed member, the second fixed member, the first movable member, and the second movable member. The first fixed member is disposed relative to the first movable member, and the second fixed member is disposed relative to the second movable member. Thus, the dimensions of the accommodating space in the first direction and the second direction can be changed by moving the first movable member and the second movable member, so as to achieve changeover of the material bin.

[0032] In some embodiments, a moving distance of the first movable member along the first direction is L_1 , and $0\text{ mm} \leq L_1 \leq 260\text{ mm}$.

[0033] A variety of types of double-sided adhesive tapes can be accommodated by controlling the moving distance of the first movable member along the first direction to be L_1 , and a spacing between the material bins of a same material zone can be appropriate, allowing the structure of the material bin apparatus to be compact.

[0034] In some embodiments, the moving distance of the first movable member along the first direction is L_1 , and $25\text{ mm} \leq L_1 \leq 140\text{ mm}$.

[0035] Thus, a variety of types of double-sided adhesive tapes can be accommodated, and the spacing between the material bins of the same material zone can be appropriate, allowing the structure of the material bin apparatus to be more compact.

[0036] In some embodiments, a moving distance of the second movable member along the second direction is L_2 , and $10\text{ mm} \leq L_2 < 800\text{ mm}$.

[0037] A variety of types of double-sided adhesive tapes can be accommodated by controlling the moving distance of the second movable member along the second direction to be L_2 , and the

spacing between the material bins of the same material zone can be appropriate, allowing the structure of the material bin apparatus to be compact.

[0038] In some embodiments, the moving distance of the second movable member along the second direction is L_2 , and $130\text{ mm} \leq L_2 \leq 305\text{ mm}$.

[0039] Thus, a variety of types of double-sided adhesive tapes can be accommodated, and the spacing between the material bins of the same material zone can be appropriate, allowing the structure of the material bin apparatus to be more compact.

[0040] According to a second aspect, this disclosure provides an adhesive application device, and the adhesive application device includes the material bin apparatuses according to any one of the foregoing embodiments.

[0041] Thus, the plurality of material bins can be configured to simultaneously feed, improving the feeding efficiency, thereby improving the production efficiency. Moreover, different types of materials (such as strip materials and plate materials) can be accommodated by adjusting the position of the movable member, thereby improving the compatibility of the material bin apparatus. In addition, the movable members of at least some of the material bins can synchronously move away from or approach the fixed members. That is, the movable members of the plurality of material bins can be simultaneously adjusted. This is conducive to achieving quick changeover, thereby improving the production efficiency.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0042] Persons of ordinary skill in the art will be clear about various other advantages and benefits by reading the detailed description of embodiments below. The accompanying drawings are only used to illustrate some embodiments and are not considered as a limitation of this disclosure. Moreover, in all the accompanying drawings, the same reference signs represent the same components. In the accompanying drawings:

[0043] FIG. 1 is a schematic structural diagram of a material bin apparatus from a first perspective according to some embodiments of this disclosure;

[0044] FIG. 2 is a schematic structural diagram of the material bin apparatus from a second perspective shown in FIG. 1;

[0045] FIG. 3 is a schematic structural diagram of the material bin apparatus from a third perspective shown in FIG. 1; and

[0046] FIG. 4 is a schematic structural diagram of the material bin apparatus from a fourth perspective shown in FIG. 1.

[0047] Reference signs are described as follows: [0048] **10**, base; **10a**, first mounting groove; **10b**, first sliding rail; **10c**, second mounting groove; **10d**, second sliding rail; **20**, material bin; **20a**, accommodating space; **21**, fixed member; **211**, first fixed member; **212**, second fixed member; **22**, movable member; **221**, first movable member; **222**, second movable member; **30**, turntable; **40**, spacer; **50**, drive mechanism; **51**, first lead screw; **52**, second lead screw; **53**, connecting member; **100**, material bin apparatus; and **100a**, material zone.

DESCRIPTION OF EMBODIMENTS

[0049] The following will describe the embodiments of the technical solutions of this disclosure in detail with reference to the accompanying drawings. The following embodiments are only used to more clearly illustrate the technical solutions of this disclosure, and are therefore only examples and cannot be used to limit the scope of protection of this disclosure.

[0050] Unless otherwise defined, all technical and scientific terms used herein shall have the same meanings as commonly understood by persons skilled in the art to which this disclosure relates. The terms used herein are intended to merely describe the specific embodiments rather than to limit

this disclosure. The terms “include”, “comprise”, and “have” and any other variations thereof of this application are intended to cover non-exclusive inclusions.

[0051] In the description of the embodiments of this disclosure, the technical terms “first”, “second”, “third”, and the like are only used to distinguish different objects, and shall not be understood as any indication or implication of relative importance or any implicit indication of the number, specific sequence, or primary-secondary relationship of the technical features indicated. In the description of the embodiments of this disclosure, “a plurality of” means more than two, unless otherwise specifically defined.

[0052] In this specification, “embodiment” means that specific features, structures, or characteristics described with reference to the embodiment may be included in at least one embodiment of this disclosure. The term “embodiment” appearing in various places in the specification does not necessarily refer to the same embodiment or an independent or alternative embodiment that is exclusive of other embodiments. It is explicitly or implicitly understood by persons skilled in the art that the embodiments described herein may be combined with other embodiments.

[0053] In the description of the embodiments of this disclosure, the term “and/or” is only an associative relationship for describing associated objects, indicating that three relationships may be present. For example, A and/or B may indicate the following three cases: presence of only A, presence of both A and B, and presence of only B. In addition, a character “/” in this specification generally indicates an “or” relationship between contextually associated objects.

[0054] In the description of the embodiments of this disclosure, the technical terms “length”, “width”, “thickness”, “upper”, “lower”, “front”, “rear”, “left”, “right”, “vertical”, “horizontal”, “top”, “bottom”, “inner”, “outer”, “circumferential”, and the like indicate orientation or positional relationships based on the orientation or positional relationships shown in the accompanying drawings, and are only for the convenience of describing the embodiments of this disclosure and simplifying the description, and are not intended to indicate or imply that the referred device or element must have a specific orientation, be constructed, operated, or used in a specific orientation. Therefore, it should not be understood as a limitation on the embodiments of this disclosure.

[0055] In the description of the embodiments of this disclosure, unless otherwise specified and defined explicitly, the technical terms “mounting”, “connecting”, “joining”, “fixing” should be understood in their general senses. For example, they may refer to a fixed connection, a detachable connection, or an integral connection, may refer to a mechanical connection or electrical connection, and may refer to a direct connection, an indirect connection via an intermediate medium, an internal communication between two elements, or an interaction between two elements. Persons of ordinary skill in the art can understand specific meanings of the foregoing terms in the embodiments of this disclosure as appropriate to specific situations.

[0056] In the description of the embodiments of this disclosure, unless otherwise specified and defined explicitly, the technical term “contact” should be understood in a general sense, which can be direct contact or contact through an intermediate medium layer, can be contact with essentially no interaction force between the two contacting parties, or can be contact with interaction force between the two contacting parties.

[0057] The following provides a detailed description of this disclosure.

[0058] Currently, the application of new energy batteries in both daily life and industry is becoming increasingly widespread. New energy batteries have been widely used in energy storage power supply systems such as hydroelectric power plants, thermal power plants, wind power plants, and solar power plants, as well as many other fields including electric transportation tools such as electric bicycles, electric motorcycles, and electric vehicles, and aerospace.

[0059] In the embodiments of this disclosure, the battery can be a battery cell. A battery cell refers to a basic unit that can achieve mutual conversion between chemical energy and electrical energy, and can be configured to make battery modules or battery packs, thereby supplying power to

electrical devices. The battery cell can be a secondary battery, and the secondary battery refers to a battery cell whose active material can be activated for continuous use through charging after the battery is discharged. The battery cell may be a lithium-ion battery, a sodium-ion battery, a sodium-lithium-ion battery, a lithium metal battery, a sodium metal battery, a lithium-sulfur battery, a magnesium-ion battery, a nickel-hydrogen battery, a nickel-cadmium battery, a lead storage battery, and the like. This is not limited in the embodiments of this disclosure.

[0060] In the embodiments of this disclosure, the battery can also be a battery module or battery pack including one or more battery cells.

[0061] In the production of the battery pack, an automatic adhesive application machine can be used to attach a double-sided adhesive tape to a surface of the battery cell. During the production process, a material bin apparatus configured to store a double-sided adhesive tape cannot accommodate a variety of types of double-sided adhesive tapes, resulting in long tooling changeover times and reduced production efficiency.

[0062] Based on the foregoing considerations, this disclosure designs a material bin apparatus and an adhesive application device, where the material bin apparatus includes a base and a plurality of material bins. The material bin is configured to include a fixed member and a movable member. The movable member is movably disposed on the base and define an accommodating space with the fixed member, where at least some of the movable members of the material bins can synchronously move away from or approach the fixed members. Thus, the plurality of material bins can be configured to simultaneously feed, improving the feeding efficiency, thereby improving the production efficiency. Moreover, different types of materials (such as strip materials and plate materials) can be accommodated by adjusting the position of the movable member, improving the compatibility of the material bin apparatus, thereby reducing the cost of the material bin apparatus. In addition, the movable members of at least some of the material bins can synchronously move away from or approach the fixed members. That is, the movable members of the plurality of material bins can be simultaneously adjusted. This is conducive to achieving quick changeover, thereby improving the production efficiency.

[0063] Embodiments of this disclosure provide an adhesive application device. The adhesive application device includes the material bin apparatus **100**.

[0064] The material bin apparatus **100** is configured to store materials.

[0065] The specific type of the material is not limited herein, and the embodiments of this disclosure are described by taking an example in which the material is a double-sided adhesive tape.

[0066] The adhesive application device includes a film tearing device, and the film tearing device can be configured to tear away release paper from at least one side of the double-sided adhesive tape.

[0067] The adhesive application device includes a conveying system, and the conveying system can be configured to convey the double-sided adhesive tape to the film tearing device.

[0068] In some embodiments of this disclosure, referring to FIGS. **1** to **4**, the material bin apparatus **100** includes a base **10** and a plurality of material bins **20**. An accommodating space **20a** is formed in each material bin **20**, and the accommodating space **20a** is configured for stacking and storing a plurality of materials. The material bin **20** includes a fixed member **21** and a movable member **22**. The fixed member **21** is fixed to the base **10**. The movable member **22** is movably disposed on the base **10** and defines the accommodating space **20a** with the fixed member **21**. The movable members **22** of at least some of the material bins **20** can synchronously move away from or approach the fixed members **21**.

[0069] The accommodating space **20a** is formed in each material bin **20**, and the accommodating space **20a** is configured for stacking and storing the plurality of materials. That is, the plurality of materials are stacked within the accommodating space **20a**, and an opening is formed at an upper end of the accommodating space **20a**, allowing the plurality of materials to be sequentially taken

out from the opening at the upper end of the accommodating space **20a**.

[0070] The material bin **20** includes the fixed member **21** and the movable member **22**. The movable member **22** defines the accommodating space **20a** with the fixed member **21**.

[0071] The fixed member **21** is fixed to the base **10**.

[0072] The specific method of fixing the fixed member **21** to the base **10** is not limited herein. For example, the fixed member **21** can be connected to the base **10** through snap-fit connection, fastening connection, plug-in connection, or the like.

[0073] The fastening connection includes but is not limited to screw connection, bolt connection, rivet connection, or the like.

[0074] The movable member **22** is movably disposed on the base **10** and defines the accommodating space **20a** with the fixed member **21**. That is, a distance between the movable member **22** and the fixed member **21** can be controlled by adjusting a position of the movable member **22**, so as to accommodate different types of materials (such as strip materials and plate materials), thereby improving the compatibility of the material bin apparatus **100**.

[0075] Thus, the fixed member **21** can be used as a reference standard. The movable member **22** moves to control the distance from the fixed member **21** and defines the accommodating space **20a** with the fixed member **21**, where the accommodating space **20a** is suitable for different types of double-sided adhesive tapes.

[0076] The fixed member **21** and the movable member **22** extend along a height direction, allowing the movable member **22** and the fixed member **21** to define the accommodating space **20a** extending along the height direction, thereby allowing the accommodating space **20a** to stack and store the plurality of materials.

[0077] The movable members **22** of at least some of the material bins **20** can synchronously move away from or approach the fixed members **21**. That is, the movable members **22** of the plurality of material bins **20** can synchronously approach or move away from the fixed members **21** by simultaneously adjusting the movable members **22**, achieving quick and simultaneous adjustment of the plurality of material bins **20**, thereby achieving quick changeover.

[0078] The material bin apparatus **100** provided by the embodiments of this disclosure includes the base **10** and the plurality of material bins **20**. The material bin **20** includes the fixed member **21** and the movable member **22**. The movable member **22** is movably disposed on the base **10** and defines the accommodating space **20a** with the fixed member **21**, where the movable members **22** of at least some of the material bins **20** can synchronously move away from or approach the fixed members **21**. Thus, the plurality of material bins **20** can be simultaneously configured to feed, improving the feeding efficiency, thereby improving the production efficiency. Moreover, different types of materials (such as strip materials and plate materials) can be accommodated by adjusting the position of the movable member **22**, improving the compatibility of the material bin apparatus **100**, thereby reducing the cost of the material bin apparatus **100**. In addition, the movable members **22** of at least some of the material bins **20** can synchronously move away from or approach the fixed members **21**, that is, the movable members **22** of the plurality of material bins **20** can be simultaneously adjusted. This is conducive to achieving quick changeover, thereby improving the production efficiency.

[0079] In some embodiments, referring to FIGS. 1 and 2, the material bin apparatus **100** includes a plurality of material zones **100a**. Each material zone **100a** includes the plurality of material bins **20**, and each material zone **100a** has a feeding state and a preparation state.

[0080] The material bin apparatus **100** includes the plurality of material zones **100a**. The plurality of material zones **100a** are provided. When one of the material zones **100a** is subjected to feeding, the other material zones **100a** can be configured for material preparation, thus allowing for feeding without shutdown and material change without shutdown, thereby improving the operating efficiency.

[0081] The plurality of material zones **100a** are provided, so that manual material change can be

performed on the material bins **20** of one material zone **100a**, and the material bins **20** of the other material zones **100a** can still operate normally, thus avoiding the problem of waiting for material change and improving the processing efficiency.

[0082] Each material zone **100a** has the feeding state and the preparation state, where the feeding state of the material zone **100a** refers to that the material zone **100a** is not subjected to feeding, that is, the material zone **100a** is in use. The preparation state of the material zone **100a** refers to that the material zone **100a** is subjected to feeding, that is, the material zone **100a** is not in use, and the material zone **100a** may be in the preparation state during material supplement or after material supplement.

[0083] The material bin apparatus **100** is provided with the plurality of material zones **100a**. During the use of the material bin apparatus **100**, when the material in the feeding state of the material zone **100a** is used up, the material zone **100a** in the feeding state is switched to the preparation state for material supplement, and the material zone **100a** in the preparation state is synchronously switched to the feeding state, thus allowing for feeding without shutdown and material change without shutdown, avoiding the problem of waiting for material change, and improving the processing efficiency.

[0084] Referring to FIGS. **1** to **4**, each material zone **100a** includes the plurality of material bins **20**, the plurality of material bins **20** can be simultaneously used for feeding, improving the feeding efficiency, thereby improving the production efficiency.

[0085] In a specific embodiment, referring to FIGS. **1** to **3**, the material bin apparatus **100** includes two material zones **100a**, with one material zone **100a** being in use and the other in preparation, that is, during the use of the material bin apparatus **100**, when one material zone **100a** is in the feeding state, the other one is in the preparation state. When the material in the feeding state of the material zone **100a** is used up, the material zone **100a** in the feeding state is switched to the preparation state for material supplement, and the material zone **100a** in the preparation state is synchronously switched to the feeding state, thus allowing for feeding without shutdown and material change without shutdown. The material bin apparatus **100** is provided with two material zones **100a**, allowing for feeding without shutdown and material change without shutdown, and allowing the structure of the material bin apparatus **100** to be compact and have reduced space occupation.

[0086] Certainly, in other embodiments, the number of the material zones **100a** in the material bin apparatus **100** may be three, four, five, or more.

[0087] It should be noted that the specific method of switching each material zone **100a** between the feeding state and the preparation state is not limited herein.

[0088] In some embodiments, the base **10** can rotate to switch each material zone **100a** between the feeding state and the preparation state. That is, the base **10** is controlled to rotate so as to switch each material zone **100a** between the feeding state and the preparation state. The structure for rotation of the base **10** is simple, and a control method is simple and efficient.

[0089] It should be noted that a specific method for controlling the rotation of the base **10** is not limited herein.

[0090] In some embodiments, referring to FIGS. **1** to **4**, the material bin apparatus **100** includes a turntable **30**. The base **10** is disposed on the turntable **30**. The turntable **30** rotates to drive the base **10** to rotate.

[0091] That is, turntable **30** is provided, and the base **10** is fixed to the turntable **30**. The turntable **30** is rotated to drive such that the turntable **30** drives the base **10** to rotate. This method achieves the rotation of the base **10** without affecting the formation of the material bin **20** on the base **10**, featuring a simple structure.

[0092] The specific method of depositing the base **10** on the turntable **30** is not limited herein. For example, the base **10** and the turntable **30** can be connected through magnetic connection, snap-fit connection, fastening connection, plug-in connection, or the like.

[0093] The specific shape of the turntable **30** is not limited herein and may be, for example, a circular shape, a polygonal shape, an elliptical shape, a track shape, an irregular shape, or the like.

[0094] The track shape refers to a shape similar to an athletic track and is formed by two semicircles and two parallel straight sides that are alternately connected.

[0095] The turntable **30** in the embodiments of this disclosure is circular. In a case of the same surface area, the circular turntable **30** requires the minimum rotation space, which can effectively save space, enabling the structure of the material bin apparatus **100** to be more compact.

[0096] The specific shape of the base **10** is not limited herein and may be, for example, a circular shape, a polygonal shape, an elliptical shape, a track shape, an irregular shape, or the like.

[0097] The base **10** in the embodiments of this disclosure is square. The square base **10** can fully utilize space to provide the material bin **20**.

[0098] It should be noted that a specific method for controlling the rotation of the turntable **30** is not limited herein. For example, the rotation of the turntable **30** can be controlled by a servo motor.

[0099] In some other embodiments, the material bin apparatus **100** may alternatively not include the turntable **30**, that is, the base **10** can be directly controlled to rotate, for example, the rotation of the base **10** can be controlled by the servo motor.

[0100] In some embodiments, referring to FIGS. **1** to **4**, the material bin apparatus **100** includes a spacer **40** disposed on the base **10**. Two material zones **100a** are provided. Moreover, the two material zones **100a** are symmetrically disposed on two sides of the spacer **40**.

[0101] The material bin apparatus **100** includes the spacer **40** disposed on the base **10**. The spacer **40** can be configured to separate different material zones **100a**. This is conducive to avoiding the influence between adjacent material zones **100a** to some extent, and ensuring the reliability of feeding and preparation in the material bin apparatus **100**. In addition, the spacer **40** can also be used as a positioning reference or for orientation, improving the convenience of using the material bin apparatus **100**.

[0102] Two material zones **100a** are provided. Moreover, the two material zones **100a** are symmetrically disposed on two sides of the spacer **40**. The spacer **40** is configured to separate the two material zones **100a**, with one material zone **100a** being in use and the other in preparation. The provision of the spacer **40** is conducive to avoiding the influence between feeding and preparation to some extent.

[0103] In some other embodiments, when more than two material zones **100a** are provided, a plurality of spacers **40** may alternatively be provided, and the plurality of material zones **100a** are separated respectively.

[0104] In some embodiments, referring to FIGS. **1** to **4**, the material bins **20** of each material zone **100a** are arranged in a straight line. The movable member **22** includes a first movable member **221**. The material bin apparatus **100** includes a drive mechanism **50**. The drive mechanism **50** can be configured to synchronously drive first movable members **221** of each material zone **100a** to reciprocate along a first direction.

[0105] The material bins **20** of each material zone **100a** are arranged in a straight line. This is conducive to simultaneously controlling the quick movement of the material bins **20**, thereby achieving quick changeover of the material bins **20** so as to be suitable for different types of double-sided adhesive tapes.

[0106] Certainly, in other embodiments, the material bins **20** can alternatively be arranged in a ring.

[0107] The number of the movable members **22** is not limited herein. For example, the movable member **22** includes the first movable member **221**.

[0108] The specific type of the first movable member **221** is not limited herein. For example, the first movable member **221** is plate-shaped and extends along the height direction.

[0109] The material bin apparatus **100** includes the drive mechanism **50**, and the drive mechanism **50** can be configured to drive the first movable member **221** to move.

[0110] The drive mechanism **50** can be configured to synchronously drive the first movable

members **221** of each material zone **100a** to reciprocate along the first direction. That is, the drive mechanism **50** can be configured to synchronously drive all the first movable members **221** of each material zone **100a** to reciprocate along the first direction. Thus, the movement of all the first movable members **221** of each material zone **100a** can be simultaneously controlled through the drive mechanism **50**, so as to achieve the simultaneous changeover of all the material bins **20** of each material zone **100a**, improving the changeover efficiency, and allowing for minute-level quick changeover, thereby improving the production efficiency.

[0111] It should be noted that the specific type of the first direction is not limited herein. For example, the first direction is an arrangement direction of the material bins **20** in each of the material zones **100a**. Thus, this is conducive to simultaneously controlling all the first movable members **221** of each material zone **100a** to move along the first direction.

[0112] Thus, this is conducive to controlling all the first movable members **221** of each material zone **100a**.

[0113] In some embodiments, referring to FIG. **1**, the movable member **22** includes a second movable member **222**. The drive mechanism **50** can be configured to synchronously drive the second movable members **222** of each material zone **100a** to reciprocate along a second direction. The second direction intersects the first direction.

[0114] The number of the movable members **22** is not limited herein. For example, the movable member **22** includes the second movable member **222**.

[0115] The specific type of the second movable member **222** is not limited herein. For example, the second movable member **222** is plate-shaped and extends along the height direction.

[0116] The material bin apparatus **100** includes the drive mechanism **50**, and the drive mechanism **50** can be configured to drive the second movable member **222** to move.

[0117] The drive mechanism **50** can be configured to synchronously drive the second movable members **222** of each material zone **100a** to reciprocate along the second direction. That is, the drive mechanism **50** can be configured to synchronously drive all the second movable members **222** of each material zone **100a** to reciprocate along the second direction. Thus, the movement of all the second movable members **222** of each material zone **100a** can be simultaneously controlled through the drive mechanism **50**, so as to achieve the simultaneous changeover of all the material bins **20** of each material zone **100a**, improving the changeover efficiency, and allowing for minute-level quick changeover, thereby improving the production efficiency.

[0118] It should be noted that the specific type of the second direction is not limited herein. For example, the second direction intersects the first direction.

[0119] In a specific embodiment, the second direction is perpendicular to the first direction.

[0120] The movable member **22** includes the first movable member **221** and the second movable member **222**. The drive mechanism **50** can synchronously drive all the first movable members **221** of each material zone **110a** to reciprocate along the first direction and can synchronously drive the second movable members **222** of each material zone **100a** to reciprocate along the second direction, so as to adjust the accommodating space **20a** of the material bin **20** from a plurality of directions, further improving the compatibility of the material bin apparatus **100**.

[0121] Certainly, in other embodiments, the number of the movable members **22** may alternatively be more than two.

[0122] It should be noted that the specific type of the drive mechanism **50** is not limited herein. For example, the drive mechanism **50** may be an electric cylinder, a screw mechanism, a motor sprocket structure, or the like.

[0123] In some embodiments, referring to FIGS. **1** to **3**, the drive mechanism **50** includes a first lead screw **51** and a plurality of first nuts mating with the first lead screw **51**. A first mounting groove **10a** and first sliding rail **10b** extending along the first direction are provided on the base **10**. The first lead screw **51** is rotatably disposed in the first mounting groove **10a**. The first nut is fixed to the bottom of each first movable member **221**. The first lead screw **51** rotates to drive each first

nut to reciprocate along the first direction. Each first nut drives each first movable member **221** to reciprocate along the first sliding rail **10b**.

[0124] The drive mechanism **50** includes the first lead screw **51** and the plurality of first nuts mating with the first lead screw **51**. That is, the drive mechanism **50** achieves transmission mainly through the mating of the first lead screw **51** and the first nuts.

[0125] The first mounting groove **10a** and first sliding rail **10b** extending along the first direction are provided on the base **10**. The first lead screw **51** is rotatably disposed in the first mounting groove **10a**. The first mounting groove **10a** is configured to accommodate the first lead screw **51**. The first sliding rail **10b** is configured to stop the rotation of each first movable member **221** and to allow each first movable member **221** to move along an extension direction of the first sliding rail **10b**.

[0126] Since the first mounting groove **10a** and the first sliding rail **10b** extend along the first direction, the first lead screw **51** also extends along the first direction.

[0127] The first nut is fixed to the bottom of each first movable member **221**. The first lead screw **51** rotates to drive each first nut to reciprocate along the first direction. Each first nut drives each first movable member **221** to reciprocate along the first sliding rail **10b**. Since the first nut is fixed to the first movable member **221**, the first nut cannot rotate. Thus, when the first lead screw **51** rotates, the first nut can move along the first lead screw **51**, thereby driving each first movable member **221** to move along the first lead screw **51**.

[0128] That is, the first nut is fixed to the bottom of each first movable member **221**, so that each first movable member **221** is rotatably connected to the first lead screw **51** through the first nut. When the first lead screw **51** rotates, the first nut can move along the first lead screw **51**, thereby driving each first movable member **221** to move along the first lead screw **51**, that is, driving each first movable member **221** to move along the extension direction of the first sliding rail **10b**. One first lead screw **51** is provided, each first movable member **221** is connected and mated with the first lead screw **51** through the first nut, thereby controlling the movement of all the first movable members **221** of each material zone **100a**, achieving simultaneously changeover of all the material bins **20** in each material zone **100a**, improving the changeover efficiency, and allowing for minute-level quick changeover, thereby improving the production efficiency. The structure of the drive mechanism **50** is simple and reliable.

[0129] It should be noted that the rotation of the first lead screw **51** can be manually controlled or controlled by the servo motor, which is not limited herein.

[0130] The first lead screw **51** can extend beyond an end of the base **10**, and an upper surface of the base **10** is higher than an upper surface of the turntable **30**. This is conducive to manually rotating the first lead screw **51**.

[0131] In some embodiments, referring to FIGS. **1** to **3**, the drive mechanism **50** includes a second lead screw **52**, a connecting member **53**, and a second nut mating with the second lead screw **52**. A second mounting groove **10c** and second sliding rail **10d** extending along the second direction are provided on the base **10**. The second lead screw **52** is rotatably disposed in the second mounting groove **10c**. The bottom of each second movable member **222** and the second nut are both fixed on the connecting member **53**. The second lead screw **52** rotates to drive the second nut to reciprocate along the second direction. This allows the connecting member **53** to drive each second movable member **222** to reciprocate along the second sliding rail **10d**.

[0132] The drive mechanism **50** includes the second lead screw **52**, the connecting member **53**, and the second nut mating with the second lead screw **52**. That is, the drive mechanism **50** achieves transmission mainly through the mating of the second lead screw **52** and the second nut.

[0133] The second mounting groove **10c** and second sliding rail **10d** extending along the second direction are provided on the base **10**. The second lead screw **52** is rotatably disposed in the second mounting groove **10c**. The second mounting groove **10c** is configured to accommodate the second lead screw **52**. The second sliding rail **10d** is configured to stop the rotation of each second

movable member **222** and the connecting member **53**, and to allow each second movable member **222** to move along an extension direction of the second sliding rail **10d**.

[0134] Since the second mounting groove **10c** and second sliding rail **10d** extend along the second direction, the second lead screw **52** also extends along the second direction.

[0135] The bottom of each second movable member **222** and the second nut are both fixed on the connecting member **53**. The second lead screw **52** rotates to drive each second nut to reciprocate along the first direction. The second lead screw **52** rotates to drive the second nut to reciprocate along the second direction. This allows the connecting member **53** to drive each second movable member **222** to reciprocate along the second sliding rail **10d**. Since the second nut is fixed to the connecting member **53**, and the second sliding rail **10d** forms an anti-rotation coupling with the connecting member **53**, the second nut cannot rotate. Therefore, when the second lead screw **52** rotates, the second nut can move along the second lead screw **52**, so as to drive the connecting member **53** to move along the second lead screw **52**, thereby allowing the connecting member **53** to drive each second movable member **222** to move along the second lead screw **52**.

[0136] One second lead screw **52** is provided, and each second movable member **222** is connected and mated with the second lead screw **52** through the second nut and the connecting member **53**, thereby controlling the movement of all the second movable members **222** of each material zone **100a**, achieving simultaneously changeover of all the material bins **20** in each material zone **100a**, improving the changeover efficiency, and allowing for minute-level quick changeover, thereby improving the production efficiency. The structure of the drive mechanism **50** is simple and reliable.

[0137] It should be noted that the rotation of the second lead screw **52** can be manually controlled or controlled by the servo motor, which is not limited herein.

[0138] In some embodiments, referring to FIG. **1**, the fixed member **21** includes a first fixed member **211** and a second fixed member **212**. The first fixed member **211** is disposed relative to the first movable member **221**. The second fixed member **212** is disposed relative to the second movable member **222**.

[0139] In the embodiments, the material bin **20** includes the first fixed member **211**, the second fixed member **212**, the first movable member **221**, and the second movable member **222**. The first fixed member **211**, the second fixed member **212**, the first movable member **221**, and the second movable member **222** define the accommodating space **20a**, that is, the double-sided adhesive tape can be limited through the first fixed member **211**, the second fixed member **212**, the first movable member **221**, and the second movable member **222**. The first fixed member **211** is disposed relative to the first movable member **221**, and the second fixed member **212** is disposed relative to the second movable member **222**. Thus, the dimensions of the accommodating space **20a** in the first direction and the second direction can be changed by moving the first movable member **221** and the second movable member **222**, so as to achieve changeover of the material bin **20**.

[0140] In some embodiments, referring to FIG. **1**, a moving distance of the first movable member **221** along the first direction is $L1$, and $0\text{ mm} < L1 \leq 260\text{ mm}$.

[0141] For example, at a temperature of 25°C ., the moving distance $L1$ of the first movable member **221** along the first direction can be measured using a vernier caliper.

[0142] For example, the moving distance $L1$ of the first movable member **221** along the first direction can be 0.1 mm, 0.5 mm, 1 mm, 1.5 mm, 2 mm, 3 mm, 4 mm, 5 mm, 8 mm, 10 mm, 20 mm, 30 mm, 41 mm, 45 mm, 76 mm, 103 mm, 120 mm, 135 mm, 146 mm, 158 mm, 163 mm, 171 mm, 186 mm, 190 mm, 200 mm, 210 mm, 222 mm, 230 mm, 241 mm, 245 mm, 250 mm, 253 mm, 257 mm, 258 mm, or 260 mm.

[0143] The moving distance of the first movable member **221** along the first direction satisfies $0\text{ mm} < L1 \leq 260\text{ mm}$. That is, a variety of types of double-sided adhesive tapes can be accommodated by controlling the moving distance of the first movable member **221** along the first direction to be $L1$, and a spacing between the material bins **20** of a same material zone **100a** can be appropriate,

allowing the structure of the material bin apparatus **100** to be compact.

[0144] In some embodiments, referring to FIG. **1**, the moving distance of the first movable member **221** along the first direction is $L1$, and $25\text{ mm} \leq L1 \leq 140\text{ mm}$. Thus, a variety of types of double-sided adhesive tapes can be accommodated, and the spacing between the material bins **20** of the same material zone **100a** can be appropriate, allowing the structure of the material bin apparatus **100** to be more compact.

[0145] In some embodiments, referring to FIG. **1**, a moving distance of the second movable member **222** along the second direction is $L2$, and $10\text{ mm} \leq L2 \leq 800\text{ mm}$.

[0146] For example, at a temperature of 25° C. , the moving distance $L2$ of the second movable member **222** along the second direction can be measured using the vernier caliper.

[0147] For example, the moving distance $L2$ of the second movable member **222** along the second direction may be 10 mm, 20 mm, 30 mm, 80 mm, 103 mm, 120 mm, 135 mm, 146 mm, 158 mm, 200 mm, 210 mm, 222 mm, 230 mm, 241 mm, 245 mm, 250 mm, 300 mm, 330 mm, 385 mm, 400 mm, 463 mm, 496 mm, 500 mm, 535 mm, 576 mm, 600 mm, 620 mm, 645 mm, 682 mm, 700 mm, 741 mm, 775 mm, 790 mm, or 800 mm.

[0148] The moving distance of the second movable member **222** along the second direction satisfies $10\text{ mm} \leq L2 \leq 800\text{ mm}$. That is, a variety of types of double-sided adhesive tapes can be accommodated by controlling the moving distance of the second movable member **222** along the second direction to be $L2$, and the spacing between the material bins **20** of the same material zone **100a** can be appropriate, allowing the structure of the material bin apparatus **100** to be compact.

[0149] In some embodiments, referring to FIG. **1**, the moving distance of the second movable member **222** along the second direction is $L2$, and $130\text{ mm} \leq L2 \leq 305\text{ mm}$. Thus, a variety of types of double-sided adhesive tapes can be accommodated, and the spacing between the material bins **20** of the same material zone **100a** can be appropriate, allowing the structure of the material bin apparatus **100** to be more compact.

[0150] In a specific embodiment, referring to FIGS. **1** to **4**, the material bin apparatus **100** includes two material zones **100a** and the spacer **40**, with one material zone **100a** being in use and the other in preparation and the material zones **100a** symmetrically disposed about the spacer **40**, that is, during the use of the material bin apparatus **100**, when one material zone **100a** is in the feeding state, the other one is in the preparation state. When the material in the feeding state of the material zone **100a** is used up, the material zone **100a** in the feeding state is switched to the preparation state for material supplement, and the material zone **100a** in the preparation state is synchronously switched to the feeding state, thus allowing for feeding without shutdown and material change without shutdown. Each material zone **100a** includes four material bins **20**, and the four material bins **20** are arranged in a straight line along the first direction. The material bin apparatus **100** includes the turntable **30**. The turntable **30** is provided, and the base **10** is fixed to the turntable **30**. The turntable **30** is rotated to drive such that the turntable **30** drives the base **10** to rotate. The material bin **20** includes the first fixed member **211**, the second fixed member **212**, the first movable member **221**, and the second movable member **222**. The first fixed member **211**, the second fixed member **212**, the first movable member **221**, and the second movable member **222** extend along the height direction and define the accommodating space **20a**. An opening is formed at the upper end of the accommodating space **20a**, allowing the plurality of materials to be sequentially taken out from the opening at the upper end of the accommodating space **20a**. That is, the double-sided adhesive tape can be limited through the first fixed member **211**, the second fixed member **212**, the first movable member **221**, and the second movable member **222**. The first fixed member **211** is disposed relative to the first movable member **221**, and the second fixed member **212** is disposed relative to the second movable member **222**. Thus, the dimensions of the accommodating space **20a** in the first direction and the second direction can be changed by moving the first movable member **221** and the second movable member **222**, so as to achieve changeover of the material bin **20**. The drive mechanism **50** includes the first lead screw **51** and the plurality of

first nuts mating with the first lead screw **51**. The first mounting groove **10a** and first sliding rail **10b** extending along the first direction are provided on the base **10**. The first lead screw **51** is rotatably disposed in the first mounting groove **10a**. The first nut is fixed to the bottom of each first movable member **221**. The first lead screw **51** rotates to drive each first nut to reciprocate along the first direction. Each first nut drives each first movable member **221** to reciprocate along the first sliding rail **10b**. The drive mechanism **50** includes the second lead screw **52**, the connecting member **53**, and the second nut mating with the second lead screw **52**. The second mounting groove **10c** and second sliding rail **10d** extending along the second direction are provided on the base **10**. The second lead screw **52** is rotatably disposed in the second mounting groove **10c**. The bottom of each second movable member **222** and the second nut are both fixed on the connecting member **53**. The second lead screw **52** rotates to drive the second nut to reciprocate along the second direction. This allows the connecting member **53** to drive each second movable member **222** to reciprocate along the second sliding rail **10d**.

[0151] The accommodating space **20a** is defined by the first fixed member **211**, the second fixed member **212**, the first movable member **221**, and the second movable member **222** and is configured to place the double-sided adhesive tape required for a battery assembly process. Four first movable members **221** are adjusted to a maximum distance in the first direction by rotating the first lead screw **51**; four second movable members **222** are adjusted to a maximum distance in the second direction by rotating the second lead screw **52**; the double-sided adhesive tape is placed; the four first movable members **221** are adjusted to the positions closely contact with the double-sided adhesive tape in the first direction by rotating the first lead screw **51**; the four second movable members **222** are adjusted to the positions closely contact with the double-sided adhesive tape in the second direction by rotating the second lead screw **52**; and the feeding of the double-sided adhesive tape is completed. When the material in the feeding state of the material zone **100a** is used up, the turntable **30** is provided, the material zone **100a** in the feeding state is switched to the preparation state for material supplement, and the material zone **100a** in the preparation state is synchronously switched to the feeding state, thus allowing for feeding without shutdown and material change without shutdown.

[0152] The foregoing embodiments are only used to illustrate the technical solutions of this disclosure and are not intended to limit it. Although this disclosure has been described in detail with reference to the foregoing embodiments, persons of ordinary skill in the art should understand that they may still make modifications to the technical solutions described in the foregoing embodiments or make equivalent replacements to some or all technical features thereof without departing from the scope of the technical solutions of the embodiments of this disclosure. They should all be covered in the scope of the claims and specification of this disclosure. In particular, as long as there is no structural conflict, the various technical features mentioned in the embodiments can be combined in any manner. This disclosure is not limited to the specific embodiments disclosed herein, but includes all technical solutions falling within the scope of this disclosure.

Industrial Applicability

[0153] The material bin apparatus provided by the embodiments of this disclosure, the plurality of material bins are configured to simultaneously feed, improving the feeding efficiency, thereby improving the production efficiency. Moreover, different types of materials (such as strip materials and plate materials) can be accommodated by adjusting a position of the movable member, thereby improving the compatibility of the material bin apparatus. In addition, the movable members of the plurality of material bins can be simultaneously adjusted. This is conducive to achieving quick changeover, thereby improving the production efficiency.

Claims

- 1.** A material bin apparatus, comprising: a base; and a plurality of material bins, wherein: an accommodating space is formed in each of the material bins, and the accommodating space is configured for stacking and storing a plurality of materials; each material bin comprises a fixed member and a movable member, wherein the fixed member is fixed to the base, and the movable member is movably disposed on the base and defines the accommodating space with the fixed member; and the movable members of at least some of the material bins are able to synchronously move away from or approach the fixed members.
- 2.** The material bin apparatus according to claim 1, wherein the material bin apparatus comprises a plurality of material zones, each of the material zones comprises a plurality of material bins, and each of the material zones has a feeding state and a preparation state.
- 3.** The material bin apparatus according to claim 2, wherein the base is able to rotate to switch each of the material zones between the feeding state and the preparation state.
- 4.** The material bin apparatus according to claim 3, further comprising: a turntable; wherein the base is disposed on the turntable, and the turntable rotates to drive the base to rotate.
- 5.** The material bin apparatus according to claim 2, further comprising: a spacer disposed on the base; wherein two material zones are provided, and the two material zones are symmetrically disposed on two sides of the spacer.
- 6.** The material bin apparatus according to claim 2, further comprising: a drive mechanism; wherein: the material bins of each of the material zones are arranged in a straight line; the movable member comprises a first movable member; and the drive mechanism is able to be configured to synchronously drive the first movable members of each of the material zones to reciprocate along a first direction.
- 7.** The material bin apparatus according to claim 6, wherein: the movable member comprises a second movable member; and the drive mechanism is able to be configured to synchronously drive the second movable members of each of the material zones to reciprocate along a second direction, the second direction intersecting the first direction; and/or the first direction is an arrangement direction of each of the material bins in each of the material zones.
- 8.** The material bin apparatus according to claim 6, wherein the drive mechanism comprises a first lead screw and a plurality of first nuts mating with the first lead screw; a first mounting groove and first sliding rail extending along the first direction are provided on the base; the first lead screw is rotatably disposed in the first mounting groove; the first nut is fixed to the bottom of each of the first movable members; the first lead screw rotates to drive each of the first nuts to reciprocate along the first direction; and each of the first nuts drives each of the first movable members to reciprocate along the first sliding rail.
- 9.** The material bin apparatus according to claim 7, wherein the drive mechanism comprises a second lead screw, a connecting member, and a second nut mating with the second lead screw; a second mounting groove and second sliding rail extending along the second direction are provided on the base; the second lead screw is rotatably disposed in the second mounting groove; the bottom of each of the second movable members and the second nut are both fixed on the connecting member; and the second lead screw rotates to drive the second nut to reciprocate along the second direction, so that the connecting member drives each of the second movable members to reciprocate along the second sliding rail.
- 10.** The material bin apparatus according to claim 7, wherein the fixed member comprises a first fixed member and a second fixed member, the first fixed member is disposed relative to the first movable member, and the second fixed member is disposed relative to the second movable member.
- 11.** The material bin apparatus according to claim 6, wherein a moving distance $L1$ of the first movable member along the first direction satisfies $0\text{ mm} \leq L1 \leq 260\text{ mm}$.
- 12.** The material bin apparatus according to claim 6, wherein a moving distance $L1$ of the first

movable member along the first direction satisfies $25\text{ mm} \leq L1 \leq 140\text{ mm}$.

13. The material bin apparatus according to claim 7, wherein a moving distance $L2$ of the second movable member along the second direction satisfies $10\text{ mm} \leq L2 \leq 800\text{ mm}$.

14. The material bin apparatus according to claim 7, wherein a moving distance $L2$ of the second movable member along the second direction satisfies $130\text{ mm} \leq L2 \leq 305\text{ mm}$.

15. An adhesive application device, comprising the material bin apparatus according to claim 1.
